



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PBL in Digital Design Verilog-HDL

Presentation

Project Title : **FPGA Implementation of Modified Error Normalized LMS
Adaptive Filter for ECG Noise Removal**

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Introduction

- Fast Fourier Transform (FFT) is widely used in signal processing applications such as biomedical signal analysis (ECG), communication systems, and image processing for frequency-domain analysis. In real-time ECG monitoring systems, FFT and adaptive filtering techniques help in identifying and removing noise components.

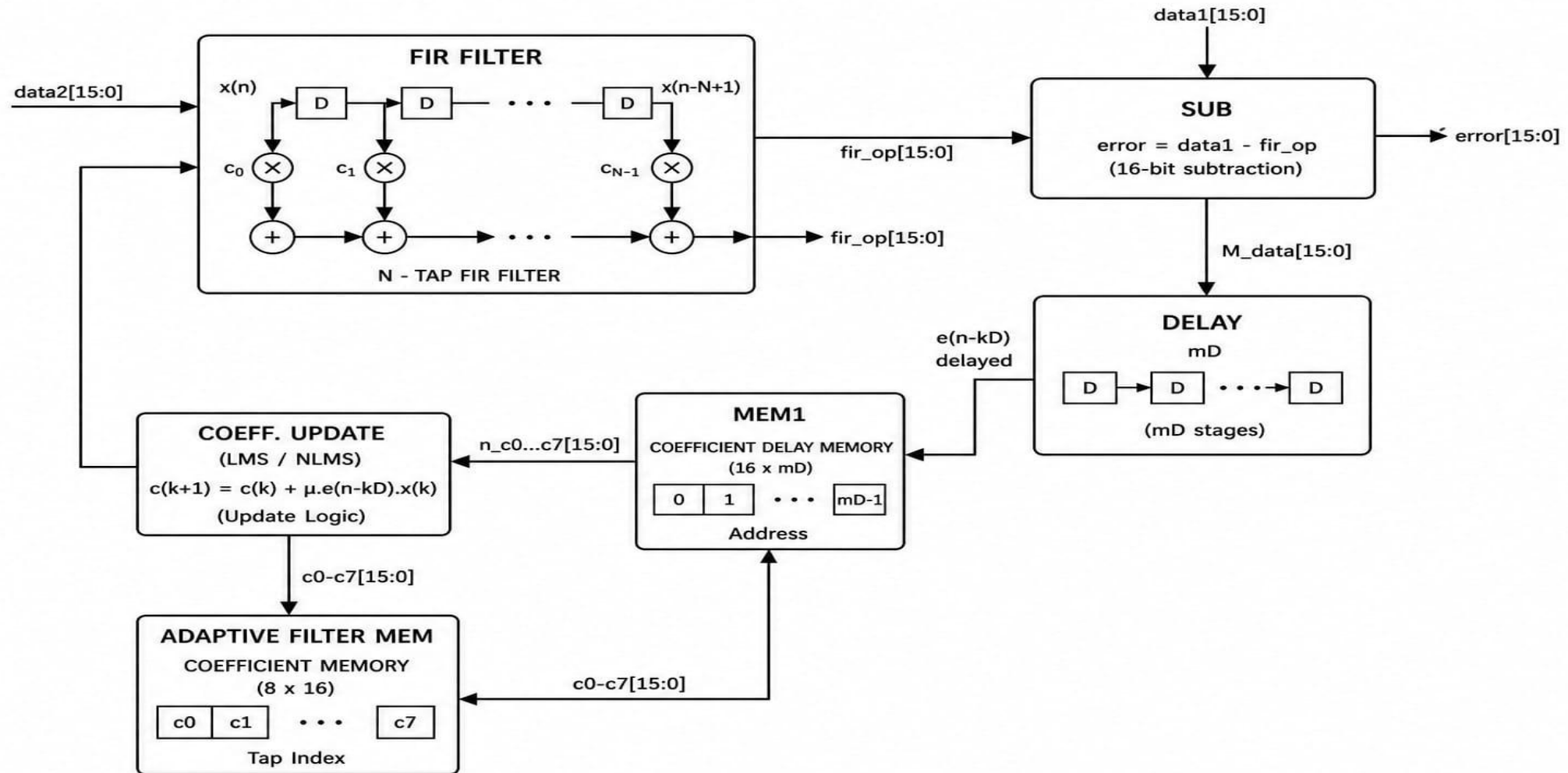
Problem Statement

- In real-time signal processing applications, FFT and adaptive filtering operations require fast and continuous computation. However, software-based FFT implementations suffer from high latency, limited throughput, and increased power consumption, making them unsuitable for real-time ECG monitoring systems.
- Additionally, direct hardware implementation of conventional algorithms may result in long critical paths, reducing processing speed. Therefore, there is a need for a hardware-efficient FPGA-based design that reduces delay, improves throughput, and supports real-time signal processing.

Objectives

- 1.To study FFT-based signal processing applications and adaptive filtering techniques for ECG noise removal.
- 2.To design a hardware-accelerated architecture suitable for real-time processing.
- 3.To implement the proposed design using Verilog HDL and perform simulation.
- 4.To deploy the design on FPGA and evaluate performance in terms of speed, latency, and power.
- 5.To compare hardware implementation with software-based approaches.
- 6.To achieve improved throughput, reduced latency, and efficient real-time operation.

Architecture Diagram



Software used

Tool	Purpose in Project	Specific Use	License/Version
ModelSim (Mentor Graphics)	Functional simulation of Verilog HDL	ECG filter behavioral simulation	Student/Academic License
Xilinx ISE Design Suite	FPGA synthesis, P&R, implementation	Virtex 5 FPGA targeting	Free WebPACK Edition
Xilinx Synthesis Tool (XST)	RTL to gate-level netlist conversion	RTL schematic generation	Included in Xilinx ISE
Virtex 5 FPGA XC5LVX330	Target FPGA chip for implementation	Logic, timing, power analysis	Physical hardware target
Microsoft Visio	Block & architecture diagram creation	Block diagram, architecture figure	Office 365 / Standalone
Verilog HDL	Hardware description language	DENLMS filter coding	IEEE 1364-2005 standard

System Design

- FIR Filter: Computes filter output $y(n)$ by multiplying input samples with weight coefficients and accumulating results
Multiplier- Accumulator (MAC) unit
N=4 taps, pipelined multiply-add
- Subtractor (SUB): Computes error signal $e(n) = d(n) - y(n)$ between desired and filter output
Combinational 16- bit subtractor
1-cycle latency
- Delay Register (mD): Introduces pipeline delay of kD cycles in error feedback path to break critical path
D flip-flop chain (register file)
kD = 1 delay stage

Multiplier (MULT): Performs signed 16x16 bit multiplication for weight update computation

Dedicated DSP48 slice

Pipelined, 2-cycle

- Adder (ADD): Performs weight accumulation: $w(n) + \mu_{en} * e * x$

Ripple-carry / carry- lookahead adder

Single-cycle

- Divider (DIV): Computes $\mu_{en} = \mu / (p + e^2)$ for error normalization

Sequential restoring divider

Multi-cycle

- Memory (MEM): Stores and retrieves current filter coefficients c0-c7

Register file (8 x 16- bit)

Synchronous read/write

- Control FSM: Generates sel_f1, sel_f2, sel_n control signals for pipeline stage selectio

- Finite State Machine (3 states)

Project distribution

Project Duration: 10 March 2026 – 2 May 2026

Task: Collection of research papers

Assigned to: All Members

Status: Completed on 2 May 2026

Task: Research paper segregation and analysis

Assigned to: Gowri & Hemashree

Status: Completed on 2 May 2026

Task: Block diagram design

Assigned to: Hemashree

Status: Completed on 2 May 2026

Task: Architecture design

Assigned to: Hemashree

Status: Completed on 2 May 2026

Task: Verilog coding

Assigned to: Evan

Status: Completed on 2 May 2026

Task: Testbench code development

Assigned to: Evan

Status: Completed on 2 May 2026

Task: Report documentation

Assigned to: Manasvi

Status: Completed on 2 May 2026

Task: PPT preparation and presentation

Assigned to: Gowri & Manasvi

Status: Completed on 2 May 2026